American Computer Science League

2023-2024 ● Contest 3: Short Problems ● Junior Division

1. Boolean Algebra

How many ordered pairs make the following Boolean expression TRUE?

$$A + B(\overline{A} + B)$$

A. 0
B. 1
C. 2
D. 3
E. 4

2. Boolean Algebra

Simplify the following Boolean expression:

$$A + \overline{B + C} + \overline{B} + \overline{\overline{A} + \overline{C}}$$

A. $\overline{A} + \overline{B}$
B. $A + B$
C. $A + \overline{B}$
D. $\overline{A}\,B$
E. $A\,\overline{B}$

3. Data Structures

Find the depth of the binary search tree for:

DODGECARAVAN

A. 2
B. 3
C. 4
D. 5
E. 6

4. Data Structures

If the following operations are done on both a stack and a queue, what is the top of the stack and the front of the queue at the end of the entire process?

```
PUSH("A")
PUSH("L")
PUSH("P")
PUSH("H")
x = POP()
x = POP()
PUSH("A")
PUSH("B")
x = POP()
PUSH("E")
PUSH("T")
x = POP()
```

A. stack top="E", queue front="H"
B. stack top="T", queue front="A"
C. stack top="T", queue front="H"
D. stack top="A", queue front="T"
E. stack top="E", queue front="A"

5. What Does This Program Do? (Arrays)

What is the output when the following program is run using the following values of array *val* given that val(0) = 3.

3	6	7	1	4	8	9	2	6	1

```
m = 1
t = 1
for x = 1 to 9
     if val(x) < val(x-1) then
          if t > m then
               m = t
          end if
          t = 1
     else
          t = t + 1
     end if
next x
output m
```

A. 1
B. 2
C. 3
D. 4
E. 5

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1. 布尔代数

请问有多少个有序对可以使得下列布尔表达式的结果为真？

$$A + B(\overline{A} + B)$$

A. 0
B. 1
C. 2
D. 3
E. 4

2. 布尔代数

请化简下列布尔表达式：

$$A + \overline{B + C} + \overline{B} + \overline{\overline{A} + \overline{C}}$$

A. $\overline{A} + \overline{B}$
B. $A + B$
C. $A + \overline{B}$
D. $\overline{A}\,B$
E. $A\,\overline{B}$

3. 数据结构

求以下字符串构成的二叉搜索树的深度：

DODGECARAVAN

A. 2
B. 3
C. 4
D. 5
E. 6

4. 数据结构

如果同时对一个栈和一个队列进行以下操作，请问在整个流程结束时栈顶和队首的元素是什么？

```
PUSH("A")
PUSH("L")
PUSH("P")
PUSH("H")
x = POP()
x = POP()
PUSH("A")
PUSH("B")
x = POP()
PUSH("E")
PUSH("T")
x = POP()
```

A. 栈顶="E", 队首="H"
B. 栈顶="T", 队首="A"
C. 栈顶="T", 队首="H"
D. 栈顶="A", 队首="T"
E. 栈顶="E", 队首="A"

5. 这个程序是做什么的？(数组)

已知 val(0) = 3, 执行下列数组 *val* , 请问此程序的输出值是多少？

3	6	7	1	4	8	9	2	6	1

```
m = 1
t = 1
for x = 1 to 9
     if val(x) < val(x-1) then
          if t > m then
               m = t
          end if
          t = 1
     else
          t = t + 1
     end if
next x
output m
```

A. 1
B. 2
C. 3
D. 4
E. 5